

DevOps Lab Documentation

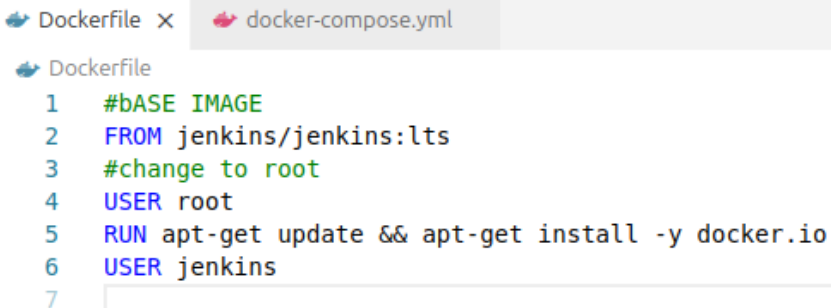
Program 5: Demonstrate CI Integration by Running Jenkins in a Docker Container

Program-5

—>Dockerfile

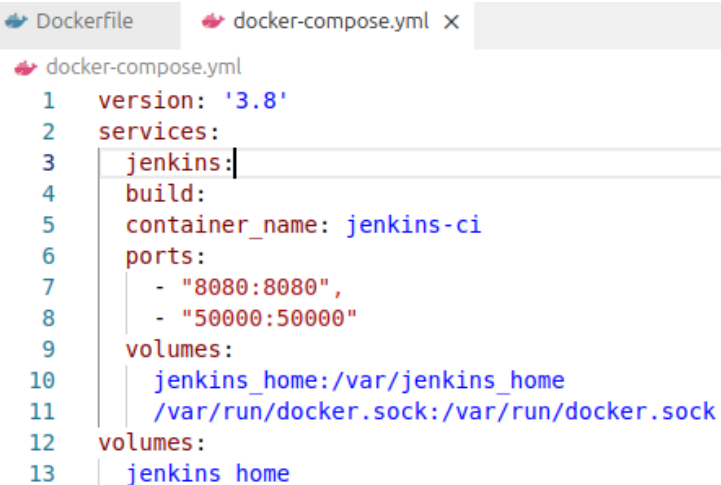
—>docker-compose.yml

S1. Create a Dockerfile



```
Dockerfile
1  #base image
2  FROM jenkins/jenkins:lts
3  #change to root
4  USER root
5  RUN apt-get update && apt-get install -y docker.io
6  USER jenkins
7
```

S2. Make a file docker-compose.yml



```
docker-compose.yml
1  version: '3.8'
2  services:
3    jenkins:
4      build:
5      container_name: jenkins-ci
6      ports:
7        - "8080:8080",
8        - "50000:50000"
9      volumes:
10       jenkins_home:/var/jenkins_home
11       /var/run/docker.sock:/var/run/docker.sock
12 volumes:
13   jenkins_home
```

S3. Build and Run Jenkins

docker build -t program-5 .

```
[+] Building 431.1s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 167B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7
=> => sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e893b800 2.58kB / 2.58kB
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa812cc605644 12.71kB / 12.71kB
=> => sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b33 49.29MB / 49.29MB
=> => sha256:5687d4d0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 65.08MB / 65.08MB
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB
=> => sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 3.12kB / 3.12kB
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627 95.32MB / 95.32MB
=> => extracting sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b33
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 189B / 189B
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fd88e9c009f6874e7221acb06245 6.29MB / 6.29MB
=> => sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bfd6899 72.05MB / 72.05MB
```

docker run -p 8080:8080 program-5

```
❖ [A^[[A^C1rv24mc113_varun@varun-srivastava:~/DevOpslab/program5$ docker run -p 8080:8080 program-4
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins home/war
2026-01-06 13:14:16.928+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-01-06 13:14:18.411+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-01-06 13:14:18.491+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T:
S
2026-01-06 13:14:19.138+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, (
2026-01-06 13:14:19.233+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-01-06 13:14:19.856+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins
2026-01-06 13:14:20.038+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$Core
AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9{STARTED}}
2026-01-06 13:14:20.057+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac{H
2026-01-06 13:14:20.080+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4{STARTI
2026-01-06 13:14:20.082+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=
2026-01-06 13:14:20.266+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.3
2026-01-06 13:14:20.391+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-01-06 13:14:20.410+0000 [id=59] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-01-06 13:14:21.599+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-01-06 13:14:21.605+0000 [id=57] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-01-06 13:14:21.606+0000 [id=46] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-01-06 13:14:21.825+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
```

S4. Test the running of Jenkins by logging into <http://localhost:8080>

